MIL-P-63225A (AR)
26 APRIL 1984
SUPERSEDING
MIL-P-63225 (AR)
17 August 1979

MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, M712
CONTROL SECTION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification established the requirements for the manufacture and acceptance of the M712 Guided Projectile Control Section, critical item.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Box, Wood, Cleated, Plywood
PPP-B-621	- Box, Wood, Nailed and Lock Corner
PPP-C-843	- Cushioning Material, Cellulosic
PPP-T-76	- Tape, Packaging, Paper (for carton sealing)

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging
MIL-D-3464	- Desiccants, Activated, Bagged, Packaged Use and Static Dehumidification
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-P-17667	- Paper, Wrapping, Chemically Neutral (Noncorrosive)
MIL-P-63224	- Projectile, Guided, 155mm, M712, Guidance
MIL-P-63226	- Projectile, Guided, 155mm, M712, Warhead
MIL-P-63227	- Projectile, Guided, 155mm, M712, Packaging for

STANDARDS

MILITARY

MIL-STD-100	- Engineering Drawing Practices
MIL-STD-105	- Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-810	- Environmental Test Methods
MIL-STD-1171	- Acceptance and Description Sheets

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US Army Armament Research and Development Center (ARDC)

9308000	- Control Section

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The Control Section, Guided Projectile M712, hereinafter referred to as the control section, is the aft section of the Projectile, Guided, Cannon Launched, M712 and interfaces with the aft end of the Warhead, Guided Projectile M712. Included in the control section are the control actuator subsystem,

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four each extendable wings and fins with separate release mechanisms, and an electrical energy source with appropriate electrical harnesses. Redundant second environment sensors, with associated electronics, are installed to provide an indication of the muzzle velocity of the projectile. All components are contained within a tubular shaped steel body, whose outer surface forms part of the outer surface of the projectile. The outer surface of the control section contains the two aft bourellets and an obturator.

3.2 Design and Construction.

3.2.1 Control Section. The control section shall be fabricated and assembled in accordance with drawings, parts lists, and other documents listed in drawing 9308000.

3.3 Characteristics. The characteristics of the control section shall be as specified herein.

3.3.1 Performance.

3.3.1.1 Electronic characteristics.

3.3.1.1.1 Electrical impedance. The direct current (DC) resistance characteristics of the control section electrical circuits shall be as shown in drawing 9308000.

3.3.1.1.2 Squib circuit impedance. The DC resistance of the control section squib circuits shall be as shown in drawing 9308000.

3.3.1.1.3 Electrical power requirements. The control section shall be capable of meeting all requirements of this specification when supplied with the following ranges of electrical power:

(a) 1.5 amperes (a) maximum at 26.0 to 37.0 volts DC (VDC)
(b) 3.0a maximum at -26.0 to -37.0 VDC
(c) 0.2a maximum at 15 $\pm$ 0.75 VDC
(d) 0.2a maximum at -15 $\pm$ 0.75 VDC.

3.3.1.1.4 Pneumatic power requirements. The control section shall require helium at 800 psi minimum for non-flight testing.

3.3.1.2 Wing operation.

3.3.1.2.1 Manual operation. The wings shall be adjusted in accordance with drawing 9308000 and there shall be no mechanical hindrance between the wings and the control housing when they are rotated in and out of the control housing.

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3.3.1.2.2 Squib operation. The wing mechanism shall unlock the wings after receipt of the wing unlock (WUN) signal. The wings shall fully deploy and lock after receipt of the wing extend (WEX) signal. When fully extended and locked against the associated adjusting screws, the angle of the leading edge of each wing as measured normal to the centerline of the projectile shall be as required by drawing 9308000.

3.3.1.2.3 Wing release mechanism. The wing release mechanism shall unlock the wings when the pressure cartridge port is exposed to the pressure requirements in accordance with drawing 9308100.

3.3.1.2.4 Wing extension mechanism. The wing extension mechanism shall fully extend and lock the wings when the pressure port is exposed to the pressure requirements in accordance with drawing 9308000.

3.3.1.3 Fin operation.

3.3.1.3.1 Fin position. The fin lock shall retain the fins in the locked position when subjected to a deploying load of 1.5 ± 0.2 pounds applied to the fin slot normal to the fin rotational axis. When disengaged by the fin retainer, and rotated to the fully extended position, the fins shall lock in the fully extended position. With the fins fully extended and resting against the crush pins, the angle between the leading edge of each fin and the outer surface of the control housing shall be 105 ± 2 degrees. When rotating in and out of the control housing, there shall be no mechanical hindrance between the fins and the control housing.

3.3.1.3.2 Shaft unlock and activation. The shaft locks shall be capable of releasing the actuator shafts with the following fin commands being applied to the actuators prior to and during actuator activation:

Fin Number	Command (VDC)
1 (net roll/yaw)	-7.0 ± 1.0
3 (net roll/yaw)	-1.5 ± 0.5
2 and 4 (pitch)	-0.2 to -0.15.

Activation time shall be 200 milliseconds (ms) or less. Activation time is measured from the time of cutter initiation signal generation (or its equivalent) to the time when fin feedback voltages indicate values within the limits of Figure 1.

3.3.1.3.3 Input signal/fin phasing. The fins shall rotate in response to input voltages to the pitch, yaw and roll command channels, as indicated in Table I. The fin rotation, clockwise

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(CW) or counterclockwise (CCW), is defined as the direction as viewed from the outboard end of the fin looking inboard. The fin number designations are shown on Figure 2.

3.3.1.3.4 Closed loop position gain. Shaft position shall be as determined by utilization of the shaft position feedback voltage. Figure 1 presents the limits for position gain in terms of feedback voltage versus input command, and includes the effects of all error sources such as adjustment tolerances, backlash, hysteresis, compliance, temperature and pressure variations and variations in output shaft torsional loading.

3.3.1.3.5 Duration. The minimum operating time of the control actuator system shall be 70 seconds with all axes operating continuously at maximum demand.

3.3.1.4 Second environment sensors (SES) logic.

3.3.1.4.1 "Go" operational requirements. When each sensor is exposed to a magnetic pulse as specified in 4.5.3.4 with a time period of 180 ± 20μS between the start of flux change to one forward mounted sensor until the start of flux change to its related aft mounted sensor, the output voltage from the SES electronics shall be 4.0 VDC or greater for a minimum of 0.7 second from the time of output voltage occurrence.

3.3.1.4.2 "No Go" operational requirements. When each sensor is exposed to a magnetic pulse as specified in 4.5.3.4 with a time period of 270μ or greater between the start of flux change to one forward mounted sensor until the start of flux change to its related aft mounted sensor, the output voltage from the SES electronics shall be no greater than 0.2 VDC.

3.3.2 Environmental conditions. The control section shall meet the requirements of 3.3.1 during and after exposure to the following environments, as specified. The requirements of 3.3.2.3, 3.3.2.4, and 3.3.2.6, apply only with the control section assembled into an all-up-round (AUR) configuration (either tactical or simulated) and with the test item installed in a tactical container (MIL-P-63227). After exposure to these environmental conditions, the control section shall show no evidence of physical damage which would render it unsuitable for its intended use.

3.3.2.1 High temperature. The control section shall meet the requirements of 3.3.1 during ambient temperature of 145°F (62.8°C).

3.3.2.2 Low temperature. The control section shall meet the requirements of 3.3.1 during ambient temperature of -25°F (-31.7°C).

3.3.2.3 Temperature shock. The control section, assembled into a test item, shall meet the requirements of 3.3.1 after exposure to temperature shock in either direction between -50°F (-45.6°C) and 160°F (71°C).

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Nominal Rated Values (Vdc)

	Y/R	Pitch
Input Command Volts	10.0	7.50
Feedback Volts	11.10	8.32

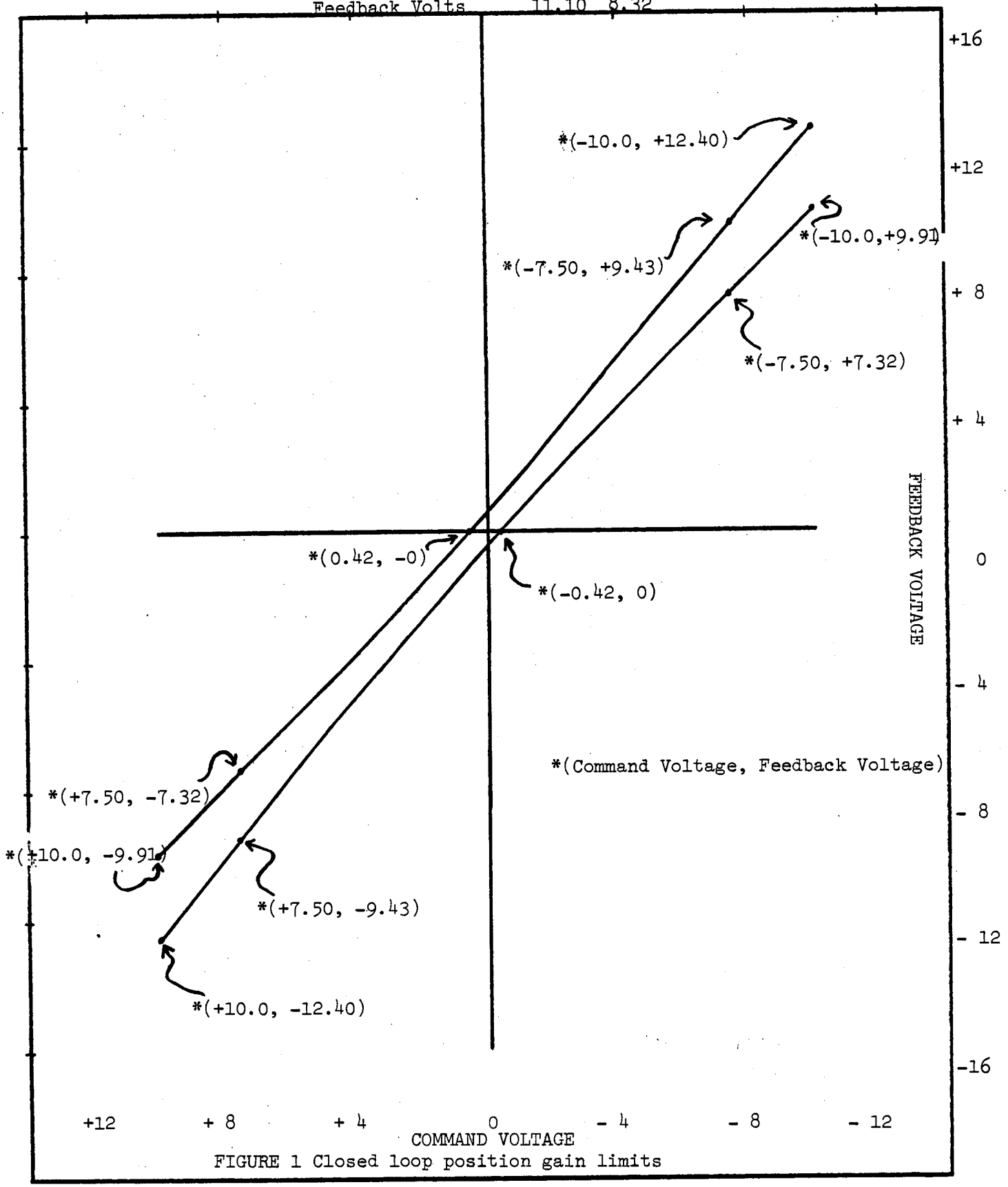


FIGURE 1 Closed loop position gain limits

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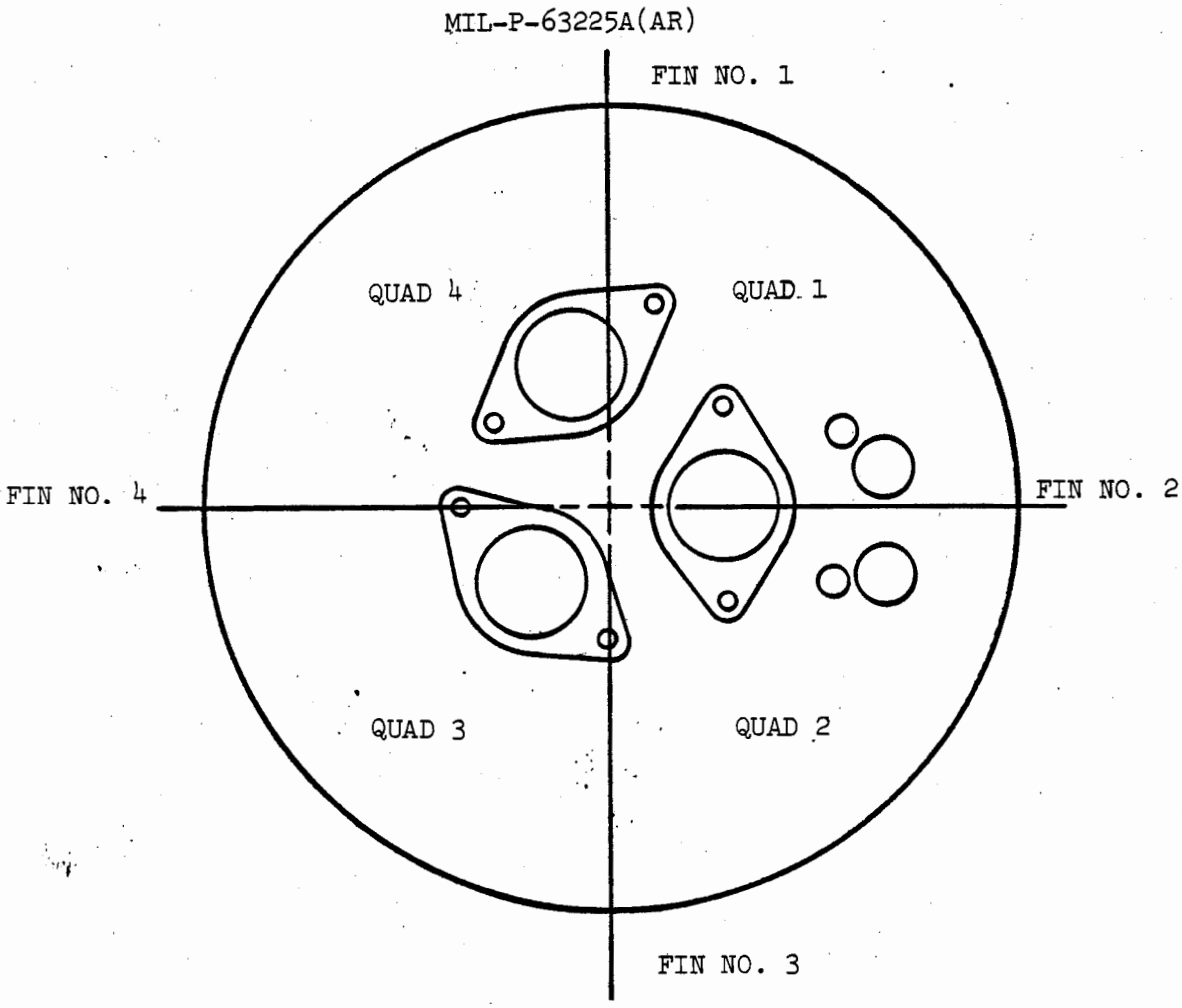


FIGURE 2 Fin designation (view looking forward)

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TABLE I Input Signal/Fin Phasing

Channel	Signal Polarity	Fin No.	Rotation
Pitch	Negative	2	CW
		4	CCW
	Positive	2	CCW
		4	CW
Yaw	Negative	1	CCW
		3	CW
	Positive	1	CW
		3	CCW
Roll	Negative	1	CCW
		3	CCW
	Positive	1	CW
		3	CW

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3.3.2.4 Transportation vibration. The control section, assembled into a test item, shall meet the requirements of 3.3.1 after exposure to the transportation vibration requirements of MIL-STD-810, Method 514.2, Procedure X.

3.3.2.5 Longitudinal axis vibration. The control section shall meet the requirements of 3.3.1 during exposure to the applicable random vibration requirements of drawing 9305340.

3.3.2.6 Fragility shock. The control section, assembled into a test item, shall meet the requirements of 3.3.1 after exposure in all three axes to half sine wave mechanical shock pulses having nominal durations of 10 ms, with nominal peak accelerations of 200g.

3.3.2.7 Launch shock. The control section shall meet the requirements of 3.3.1 after exposure to a forward longitudinal acceleration of 9000g ± 300g at temperatures ranging from 70°F (21.1°C) to 145°F (62.8°C) and 8000g ± 400g at -25°F (-31.7°C) linearly increasing to 9000g at 70°F (21.1°C). This requirement shall apply following exposure to the temperature shock of 3.3.2.3, transportation vibration of 3.3.2.4, and fragility shock of 3.3.2.6.

3.4 First article inspection. This specification makes provision for first article inspection. First article samples shall meet all the requirements of this specification. Requirements for submission of the first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts shall be fabricated and finished in a thoroughly workmanlike manner. Except as otherwise specified, the parts shall be free of cracks, scratches, burrs, dents, gouges, surface defects or any surface discontinuities. The items shall be free from dirt, chips, grease, corrosion and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and legible. Exterior surface coatings shall be continuous. Scratches not exposing base metal are permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The following types of inspection shall be conductd on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 10 control section assemblies and the parts and assemblies listed below. All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the procuring activity.

Part description	Drawings	Quantity
Aft Closure - Machining	9308015	5
Clip, Fin, Spring	9308018	5
Obturator	9308030	5
Pin-Fin Stop	9308032	5
Wing-Machining	9308035	5
Piston - Wing Extension	9308043	5
Housing, Wing Extension - Machining	9308045	5
Plug, Wing Extension	9308047	5
Control Housing	9308050	5
Stud-Pressure Vessel Support	9308051	5
Eccentric, Inner-Pressure Vessel Support	9308052	5
Eccentric, Outer-Pressure Vessel Support	9308053	5
Tube, Standpipe	9308057	5
Nut, Special - Pressure Vessel Support	9308059	5
Pin - Wing Pivot	9308067	5
Key, Control Housing	9308069	5
Spider, Wing Extension - Machining	9308072	5
Cable Assembly, Branched, W1 Control Section	9308076	5
Spring - Fin Retention	9308080	5
Spring - Lock, Spider	9308081	5

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Wing Release Mechanism Assy.	9308100	5
Mount, Wing Release Machining	9308104	5
Piston, Actuator	9308113	5
Guide, Wing Release, Machining	9308119	5
Rotor, Wing Release, Machining	9308120	5
Mount Actuator - Machining	9308124	5
Mount Assembly - Wing Release - Welded	9308125	5

4.3.2 Inspections to be performed. The first article test samples shall be subjected by the procuring activity to the examinations and tests in accordance with Table II. Also the first article test samples may be subjected by the procuring activity to any or all of the examinations and tests specified in this specification and to any or all requirements of the applicable drawings. Environmental tests shall be conducted in the sequence indicated. The testing shall be considered destructive and the tested assemblies shall not be returned to a lot for delivery.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article samples shall be rejected. The procuring activity reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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Table II. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Control Section		9308000
			NEXT HIGHER ASSEMBLY 9305300

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electrical impedance	10	-	3.3.1.1.1	4.5.3.1
	Squib circuit impendance	10	-	3.3.1.1.2	4.5.3.2
	Electrical power requirements	10	-	3.3.1.1.3	4.5.3.3
	Second environment sensors logic	10	-	3.3.1.4	4.5.3.4
	Manual operation (wings)	10	-	3.3.1.2.1	4.5.3.5.1
	Fin position	10	-	3.3.1.3.1	4.5.3.6.1, 4.5.3.6.2
	Input signal/fin phasing	10	-	3.3.1.3.3	4.5.3.6.3
	Closed loop position gain	10	-	3.3.1.3.4	4.5.3.6.4
	Temperature shock	10	-	3.3.2.3	4.5.4.1
	Transportation vibration	10	-	3.3.2.4	4.5.4.2
	Fragility shock	10	-	3.3.2.6	4.5.4.3

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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Table II. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63225A(AR)

PARAGRAPH	TITLE: Control Section	SHEET 2 OF 2	DRAWING NUMBER: 9308000
			NEXT HIGHER ASSEMBLY: 9305300

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group I-5 Assemblies (see note 1)				
	Launch shock-high temperature	5	-	3.3.2.7	4.5.4.4
	High temperature		-	3.3.2.1	4.5.4.6
	Squib operation	5	-	3.3.1.2.2	4.5.3.5.2
	Shaft unlock/duration cycle test	1	-	3.3.1.3.2 3.3.1.3.5	4.5.3.6.5
	Fin commands	2	-	3.3.1.3.4	4.5.3.6.6
	Functional tests	2	-	3.3.1.3.3	4.5.3.6.7
	Group II-5 Assemblies (see note 1)				
	Launch shock-low temperature	5	-	3.3.2.7	4.5.4.5
	Low temperature		-	3.3.2.2	4.5.4.6
	Squib operation	5	-	3.3.1.2.2	4.5.3.5.2
	Shaft unlock/duration cycle test	1	-	3.3.1.3.2 3.3.1.3.5	4.5.3.6.5
	Fin commands	2	-	3.3.1.3.4	4.5.3.6.6
	Functional tests	2	-	3.3.1.3.3	4.5.3.6.7

NOTES:
1. Ten assemblies shall be tested in the sequence shown for items on sheet 1 of 2. Five of these assemblies shall then be allocated to Group I for testing. The remaining 5 assemblies shall be allocated to Group II for testing .

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4.4 Quality conformance inspection.

4.4.1 Inspection lot. Unless otherwise specified herein or by the procuring activity, inspection lot size, lot formation, and presentation of lots shall be in accordance with the inspection and sampling procedures herein and the "submission of product" provisions of MIL-STD-105. Lot size shall be no greater than 1 month's production. Each lot of control sections shall consist of control actuator assemblies (CAS), second environmental sensors (SES), second environment sensor electronics (SESE), and "one shot" devices, each from not more than one interfix number, from one manufacturer.

4.4.2 Inspections.

a. Sampling plans. Unless otherwise specified, sampling shall be to level II of MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Aft Closure - Machining	1 OF 3	9308015
			NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Crack		100%	Dwg	Dwg
Major					
101	Hardness after heat treat		N/A	Dwg	Vendor lot certificate
Minor					
201	Thread dimensions		0.65%	Dwg	Gage
202	True position of thread major diameter with respect to forward inner surface diameter		0.65%	Dwg	Gage
203	Outside diameter of aft shoulder		0.65%	Dwg	Gage
204	Inside diameter of aft shoulder		0.65%	Dwg	Gage
205	True position of aft shoulder I.D. and O.D. with respect to forward I.D.		0.65%	Dwg	Gage
206	Dimensions of extraction tool interface groove		0.65%	Dwg	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9308015
4.4.2.1	Aft Closure - Machining				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	True relationship between groove diameter and forward inner surface diameter		0.65%	Dwg	Gage
208	Diameter of Obturator interface surface		0.65%	Dwg	Gage
209	True relationship between obturator interface surface and forward inner surface diameter		0.65%	Dwg	Gage
210	Forward shoulder I.D.		0.65%	Dwg	Gage
211	True relationship between forward shoulder I.D. and forward inner surface diameter		0.65%	Dwg	Gage
212	Diameter of inner surface		0.65%	Dwg	Gage
213	Perpendicularity of forward shoulder with respect to forward inner surface		0.65%	Dwg	Gage
214	Distance from forward face to forward shoulder		0.65%	Dwg	Gage
215	Distance from forward face to aft shoulder		0.65%	Dwg	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9308015
4.4.2.1	Aft Closure - Machining				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
216	Thickness of closure in domed area		0.65%	Dwg	Gage
217	Finish of machined surface		0.65%	Dwg	Gage
218	Marking		0.65%	Dwg	Visual
219	Workmanship		0.65%	3.5	4.5.5

NOTES:

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QUALITY CONFORMANCE INSPECTIONS

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9308018
4.4.2.2	Clip, Fin, Spring				NEXT HIGHER ASSEMBLY 9308100
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Hardness		0.65%	Dwg	Gage
202	Lip dimension		0.65%	Dwg	Gage
203	True position of hole with respect to lip		0.65%	Dwg	Gage
204	Workmanship		0.65%	3.5	4.5.5

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Obturator				9308030
					NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Inside diameter		0.40%	Dwg	Gage
102	True relationship of I.D. with forward and aft faces		0.40%	Dwg	Gage
103	Outside diameter (maximum)		0.40%	Dwg	Gage
104	True relationship between max O.D. and forward and aft faces		0.40%	Dwg	Gage
105	Diameter at bottom of grooves		0.40%	Dwg	Gage
Minor					
201	Length of Obturator		0.65%	Dwg	Gage or Visual Stand.
202	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Pin-Fin Stop		9308032
			NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter of body		0.65%	Dwg	Gage
202	Diameter of head		0.65%	Dwg	Gage
203	True relationship between head diameter and body diameter		0.65%	Dwg	Gage
204	Overall length		0.65%	Dwg	Gage
205	Hardness		0.65%	Dwg	Gage
206	Finish		0.65%	Dwg	Gage
207	Workmanship		0.65%	3.5	4.5.5

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Wing-Machining		9308035
			NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pivot hole location		0.65%	Dwg	Gage
202	Pivot hole diameter		0.65%	Dwg	Gage
203	True position of pivot hole with respect to datum plane and center-line of wing thickness (datum plane)		0.65%	Dwg	Gage
204	True position of slot with respect to datum planes		0.65%	Dwg	Gage
205	Machined finish		0.65%	Dwg	Gage
206	Workmanship		0.65%	3.5	4.5.5
207	Tapped hole location		0.65%	Dwg	Gage
208	True position of tapped hole with respect to datum		0.65%	Dwg	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Piston - Wing Extension		9308043
			NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
201	Hardness after heat treat		0.65%	Dwg	Gage
202	Outside Diameter of piston		0.65%	Dwg	Gage
203	Diameter of "0" ring groove		0.65%	Dwg	Gage
204	Runoutof "0" ring groove diameter with respect to piston diameter		0.65%	Dwg	Gage
205	Shaft diameter		0.65%	Dwg	Gage
206	True relationship between shaft diameter and piston diameter		0.65%	Dwg	Gage
207	Depth of "0" ring groove		0.65%	Dwg	Gage
208	Width of "0" ring groove		0.65%	Dwg	Gage
209	Machined finish		0.65%	Dwg	Gage
210	Workmanship		0.65%	3.5	4.5.5

NOTES:

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.7	Housing, Wing Extension - Machining	1	3	9308045
				NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter of mounting shoulder		0.65%	Dwg	Gage
202	Outside diameter of housing		0.65%	Dwg	Gage
203	True relationship between O.D. and reference diameter		0.65%	Dwg	Gage
204	Inside diameter of cylinder		0.65%	Dwg	Gage
205	True position between cylinder I.D., mounting shoulder and housing O.D.		0.65%	Dwg	Gage
206	Diameter of piston shaft hole		0.65%	Dwg	Gage
207	True position of piston shaft hole diameter and cylinder I.D.		0.65%	Dwg	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.7	Housing, Wing Extension - Machining				9308045
					NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208	Wing pivot holes diameter (four)		0.65%	Dwg	Gage
209	True position of wing pivot holes (two) with respect to mounting shoulder, housing O.D. and plane through vertical centerline		0.65%	Dwg	Gage
210	True position of wing pivot holes (two) with respect to mounting shoulder, housing O.D. and plane through horizontal centerline		0.65%	Dwg	Gage
211	Spring pin fastening holes diameter		0.65%	Dwg	Gage
212	Spring pin fastening holes location		0.65%	Dwg	Gage
213	True relationship between spring hole diameter, mounting shoulder horizontal and vertical centerline planes		0.65%	Dwg	Gage
214	Pin puller threads		0.65%	Dwg	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9308045
4.4.2.7	Housing, Wing Extension - Machining				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
215	True position of pin puller hole diameter, mounting shoulder and plane of vertical centerline		0.65%	Dwg	Gage
216	Machined finish		0.65%	Dwg	Gage or Visual Stand.
217	Workmanship		0.65%	3.5	4.5.5

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Plug, Wing Extension		9308047
			NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION. METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Distance between spanner holes		0.65%	Dwg	Gage
202	Thread dimensions		0.65%	Dwg	Gage
203	Depth of holes		0.65%	Dwg	Gage
204	Diameter of holes		0.65%	Dwg	Gage
205	Hardness after heat treat		0.65%	Dwg	Gage
206	Machined finish		0.65%	Dwg	Gage or Visual Stand.
207	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9308050
4.4.2.9	Control Housing				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Inspection for cracks		100%	Dwg	Dwg. 9300967
2	Minimum wall thickness center section of housing		100%	Dwg	Gage
3	Minimum thickness between keyway and crush pin hole		100%	Dwg	Gage
Major					
101	Control housing material		N/A	Dwg	Vendor lot certification
102	Weld		Dwg	Dwg	Dwg. 9300967
103	Hardness		100%	Dwg	Gage, Dwg. 9300967
104	Width and depth of snap ring groove		0.40%	Dwg	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9308050
4.4.2.9	Control Housing				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
105	Angle of forward face of snap ring groove		0.40%	Dwg	Gage
106	Maximum wall thickness center section of housing		100%	Dwg	Gage
107	Outside diameter		100%	Dwg	Gage
108	Two bourrelet diameters		0.40%	Dwg	Gage
109	Keyway dimensions		100%	Dwg	Gage
Minor					
201	Depth of SES shoulder		0.65%	Dwg	Gage
202	Distance from aft end to forward shoulder		0.65%	Dwg	Gage
203	Distance from aft end to aft shoulder		0.65%	Dwg	Gage
204	Run out of aft closure interface shoulder with respect to the axis of the forward and aft I.D.s		0.65%	Dwg	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9308050
4.4.2.9	Control Housing				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
205	Location of snap ring groove		0.65%	Dwg	Gage
206	True position of forward shoulder I.D. with respect to forward I.D.		0.65%	Dwg	Gage
207	Concentricity of the forward O.D. and the body I.D.		0.65%	Dwg	Gage
208	Run out of forward and aft faces with respect to the axis of the forward and aft I.D.s		0.65%	Dwg	Gage
210	Run out of aft diameter axis with respect to the axis of the forward and aft I.D.s		0.65%	Dwg	Gage
211	Wing extension housing mounting holes (4) locations and diameters		0.65%	Dwg	Gage
212	Aft closure threads		0.65%	Dwg	Gage
213	Finish of machined surface		0.65%	Dwg	Gage or Visual Stand.
214	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9308051
4.4.2.10	Stud-Pressure Vessel Support				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Hardness after heat treat		0.65%	Dwg	Gage
202	Locknut threads, pitch diameter		0.65%	Dwg	Gage
203	Length from face to end of support		0.65%	Dwg	Gage
204	Workmanship		0.65%	3.5	4.5.5

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MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.11	Eccentric, Inner-Pressure Vessel Support	1	1	9308052

NEXT HIGHER ASSEMBLY: 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of inside diameter with respect to outside diameter		0.65%	Dwg	Gage
202	Workmanship		0.65%	3.5	4.5.5

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Eccentric, Outer-Pressure Vessel Support	1 OF 1	9308053
			NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of inside diameter with respect to outside diameter		0.65%	Dwg	Gage
202	Workmanship		0.65%	3.5	4.5.5

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Tube, Standpipe				9308057
					NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Mounting thread dimensions		0.65%	Dwg	Gage
202	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9308059
4.4.2.14	Nut, Special - Pressure Vessel Support				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Hardness after heat treat		0.65%	Dwg	Gage
202	True position of inside diameter with respect to outside diameter		0.65%	Dwg	Gage
203	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Pin - Wing Pivot				9308067
					NEXT HIGHER ASSEMBLY
					9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Outside diameter of shank		0.65%	Dwg	Gage
202	Surface finish		0.65%	Dwg	Gage or Visual Stand.
203	Workmanship		0.65%	3.5	4.5.5

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9308069
4.4.2.16	Key, Control Housing				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Material		N/A	Dwg	Vendor lot certification
202	Hardness after heat treat		0.65%	Dwg	Gage
203	Width		0.65%	Dwg	Gage
204	Thickness		0.65%	Dwg	Gage
205	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9308072
4.4.2.17	Spider, Wing Extension - Machining				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Angle between arm and centerline of spring holes		0.65%	Dwg	Gage
202	Distance from centerline of spring hole to inside face		0.65%	Dwg	Gage
203	Spring hole diameter (2)		0.65%	Dwg	Gage
204	True position of spring holes with respect to inside diameter and arm		0.65%	Dwg	Gage
205	Sharpness of edge of arm		0.65%	Dwg	Gage
206	Angle of outside edge with respect to forward edge of arms		0.65%	Dwg	Gage
207	Workmanship		0.65%	3.5	4.5.5

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Cable Assembly, Branched, W1 Control Section				9308076
					NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Resistance of cable assembly		0.65%	Dwg	Gage
202	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Spring - Fin Retention				9308080
					NEXT HIGHER ASSEMBLY 9308100
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire diameter		0.40%	Dwg	Gage
102	Stress relief		Lot Sample	Dwg	Vendor lot certification
Minor					
201	Workmanship		0.65%	3.5	4.5.5

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9308081
4.4.2.20	Spring - Lock, Spider				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire diameter		0.40%	Dwg	Gage
102	Stress relief		Lot Sample	Dwg	Vendor lot certification
Minor					
201	Length of straight sections		0.65%	Dwg	Gage
202	Concentricity of straight sections		0.65%	Dwg	Gage
203	Workmanship		0.65%	3.5	4.5.5

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Wing Release Mechanism Assy.	1 OF 2	9308100
			NEXT HIGHER ASSEMBLY: 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load required to move springs aft		100%	Dwg	Gage
102	Guide installation screw properly torqued		0.40%	Dwg	Gage
103	Fin spring clip and fin spring retainer screws properly torqued		0.40%	Dwg	Gage
104	Clip spring retainer screw properly sealed and torqued		0.40%	Dwg	Gage
105	Wing release test		0.40%	3.3.1.2.3	4.5.3.5.3
Minor					
203	Rotor turns freely after primer and sealer are applied		0.65%	Dwg	Hand Operated
204	Workmanship		0.65%	3.5	4.5.5
205	Marking		0.65%	Dwg	Visual

NOTES:

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MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9308100
4.4.2.21	Wing Release Mechanism Assy.				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
206	Verify that spring in rotor is properly installed		0.65%	Dwg	Visual
207	Verify position of rotor pin and piston to note 13		0.65%	Dwg	Visual
208	Verify pressure cartridge is properly sealed and torqued		0.65%	Dwg	Visual
209	Verify shorting device is installed		0.65%	Dwg	Visual
210	Measure diameter of "O" ring to note 8		100%	Dwg	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.22	Mount, Wing Release Machining				9308104
					NEXT HIGHER ASSEMBLY 9308125
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Mounting hole locations		0.65%	Dwg	Gage
202	True position of datums three (3) mounting holes		0.65%	Dwg	Gage
203	Flatness of guide mounting face		0.65%	Dwg	Gage
204	True position of guide pin hole		0.65%	Dwg	Gage
205	True position of guide mounting screw hole with respect to Datums -E- & -B-		0.65%	Dwg	Gage
206	Tube mounting holes true location with respect to centerlines		0.65%	Dwg	Gage
207	Workmanship		0.65%	3.5	4.5.5
208	Marking		0.65%	Dwg	Gage
209	Finish		0.65%	Dwg	Gage

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Piston, Actuator	1 OF 1	9308113
			NEXT HIGHER ASSEMBLY: 9308100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of .2440 to -B-		0.65%	Dwg	Gage
202	True position of .1550 to -A-		0.65%	Dwg	Gage
203	True position of .1320 to -A-		0.65%	Dwg	Gage
204	Workmanship		0.65%	3.5	4.5.5

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTIONS

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Guide, Wing Release, Machining	1 OF 1	9308119
			NEXT HIGHER ASSEMBLY 9308100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Hardness after heat treatment		0.65%	Dwg	Gage
202	HUB I.D. -A-		0.65%	Dwg	Gage
203	True position between HUB O.D. -A-		0.65%	Dwg	Gage
204	Workmanship		0.65%	3.5	4.5.5

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Rotor, Wing Release, Machining	1 OF 1	9308120
			NEXT HIGHER ASSEMBLY 9308100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tensile strength of arms		0.40%	Dwg	Gage
102	Hardness after heat treatment		0.40%	Dwg	Gage
Minor					
201	Workmanship		0.65%	3.5	4.5.5
202	True position of .6150 to -A-		0.65%	Dwg	Gage
203	True position of .225 slot to -B-		0.65%	Dwg	Gage
204	Angularity of slot to -A- & -C-		0.65%	Dwg	Gage

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9308124
4.4.2.26	Mount Actuator - Machining				NEXT HIGHER ASSEMBLY 9308125
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of -D- to datums -A-, -B- & -C-		0.65%	Dwg	Gage
202	True position of -F- to -D-		0.65%	Dwg	Gage
203	True position of .2750		0.65%	Dwg	Gage
204	True position of .2710		0.65%	Dwg	Gage
205	Workmanship		0.65%	3.5	4.5.5

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.27	Mount Assembly - Wing Release - Welded		9308125
			NEXT HIGHER ASSEMBLY 9308100

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Magnetic particle		100%	Dwg	4.5.6
Minor					
201	Angularity of actuator to datums -B- & -C-		0.65%	Dwg	Gage
202	True position of .2710		0.65%	Dwg	Gage
203	Hardness after head treatment		0.65%	Dwg	Gage
204	Workmanship		0.65%	3.5	4.5.5

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER 9308000
4.4.2.28	Control Section				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Electrical impedance		100%	3.3.1.1.1	4.5.3.1
102	Squib circuit impedance		100%	3.3.1.1.2	4.5.3.2
103	Electrical power requirements		100%	3.3.1.1.3	4.5.3.3
104	Second environment sensors logic		100%	3.3.1.4	4.5.3.4
105	Manual operation		100%	3.3.1.2.1	4.5.3.5.1
106	Fin operation		100%	3.3.1.3.1	4.5.3.6.1, 4.5.3.6.2
107	Input signal/fin phasing		100%	3.3.1.3.3	4.5.3.6.3
108	Closed loop position gain		100%	3.3.1.3.4	4.5.3.6.4
109	Longitudinal axis vibration		100%	3.3.2.5	4.5.4.7

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER 9308000
4.4.2.28	Control Section				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110	Wires installed between the stand pipe tube and the battery, twisted four turns per inch, properly routed and fastened to battery		0.40%	Dwg	Visual
111	Tabs "A" and "B" of W-1 cable properly routed and fastened		0.40%	Dwg	Visual
112	Cable W-1 properly installed in stand pipe tube		100%	Dwg	Visual
113	Shielding on pressure cartridge cable properly terminated and bonded		0.40%	Dwg	Visual
114	Pressure cartridge cable properly fastened to structure.		0.40%	Dwg	Visual
115	Shielding on wing release cable properly terminated and insulated, wires properly fastened to structure		0.40%	Dwg	Visual
116	Wiring from SESE properly installed and fastened to structure		0.40%	Dwg	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Control Section	3 OF 5	9308000
			NEXT HIGHER ASSEMBLY: 9305300

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
117	Wires between control actuator base and aft closure properly installed to assure each wire will bear against aft closure		100%	Dwg	Visual
118	Cable restraint properly installed and bonded with tabs "A" and "B" of W-1 cable properly installed in cable restraint.		0.40%	Dwg	Visual
119	Wing extension pressure test		0.40%	3.3.1.2.3	4.5.3.5.4
Minor 201	Electronics assembly (A1 & A2) installation screws properly torqued and sealed		0.65%	Dwg	Gage
202	Wing release mechanism assembly installation		0.65%	Dwg	Gage
204	Aft closure properly installed and torqued		0.65%	Dwg	Gage
205	Wing extension piston retainer nut properly torqued and piston motion tested		0.65%	Dwg	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 4 OF 5		DRAWING NUMBER 9308000
4.4.2.28	Control Section				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
206	Thermal battery properly installed		0.65%	Dwg	Gage
207	Battery wiring properly installed		0.65%	Dwg	Gage
208	Resistors properly installed		0.65%	Dwg	Gage
209	Wings properly installed and inspect for freedom of motion of wings within notches of rotor		0.65%	Dwg	Visual
210	Wing pivot pin properly staked		0.65%	Dwg	Visual
211	Obturator properly installed		0.65%	Dwg	Gage
212	Fins and fin hubs are matched pairs		0.65%	Dwg	Visual
213	Fin properly cemented to hub		0.65%	Dwg	Visual
214	Control housing key installed		0.65%	Dwg	Visual
215	Gelatin capsule installed in wing housing extension		0.65%	Dwg	Visual
216	Wing extension plug properly installed		0.65%	Dwg	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63225A(AR)

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER 9308000
4.4.2.28	Control Section				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
217	Workmanship		0.65%	3.5	4.5.5
218	Marking, part numbers and serialization		0.65%	Dwg	Visual
219	Screws holding clamp to standpipe, bottle, and battery are properly torqued		0.65%	Dwg	Visual
220	Filler material is installed between standpipe, bottle, battery, and wing release mount		0.65%	Dwg	Visual

NOTES:

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4.5 Methods of inspection.

4.5.1 Test conditions.

4.5.1.1 Environmental conditions. Unless otherwise specified for the purpose of test, testing shall be conducted at the following conditions:

(a) Temperature: $72^{\circ} \pm 18^{\circ}F$ ($22.2^{\circ} \pm 10^{\circ}C$)

(b) Relative Humidity: 90 percent maximum

(c) Barometric Pressure: 28 to 32 inches of mercury

4.5.1.2 Test tolerances. All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances of limits specified.

4.5.1.3 Inspection equipment. The inspection equipments required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the inspection tables included herein. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (See 6.4).

4.5.1.4 Squib protection. Except as otherwise specified for the purpose of test, all squib leads shall be shorted to squib ground to prevent inadvertent destruction of the squibs during testing.

4.5.1.5 Special Test Wiring. For the purpose of conducting the live sequence tests, one 22 AWG insulated wire shall be soldered to each of the two 11 VDC battery terminals. These wires shall be extended through and secured to the control section in such a manner that they will be available for supplying +11 VDC to the projectile during the conduct of the tests.

4.5.1.6 All up round (AUR) configuration. Where the test methods require testing of the control section in an AUR configuration, it is intended that the AUR consist of a tactical guidance section (MIL-P-63224), a tactical control section (MIL-P-63225), and an inert warhead section. The inert warhead shall be identical to the tactical warhead (MIL-P-63226) with the exception that the explosive cavity shall be filled with inert material having the same specific gravity as the explosive composition and the fuze shall be an inert test fuze fabricated to the requirements of MIL-P-63226. Live sequence testing may be conducted with the sections only electrically connected as an AUR. Mechanical connection of the sections into an AUR is not required.

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4.5.2 Physical and dimensional characteristics inspection. The physical and dimensional characteristics shall be inspected to the requirements of 3.2.1.

4.5.3 Performance tests. The tests of 4.5.3.5.2 and 4.5.3.6.5 through 4.5.3.6.7 shall be conducted with the control section electrically assembled into an AUR configuration.

4.5.3.1 Control section electrical impedance test. The DC resistance characteristics of the control section wiring and electronics measured in the power off condition shall comply with the values indicated in drawing 9308000 when measurements are taken between the wires corresponding to the listed function.

4.5.3.2 Squib circuit impedance test. The DC resistance characteristics of the control section squib circuits measured in the power off condition shall comply with the values indicated in drawing 9308000 when measurements are taken between the wires corresponding to the listed functions. Shorting devices shall be removed from the squibs prior to start of any test requiring squib activation or squib impedance measurement (See 4.5.1.5). The meter used to measure squib resistance shall provide no more than 5 milliamperes to the squib under test to prevent inadvertent destruction of the squib.

4.5.3.3 Power requirements test. The control section shall meet the requirements of 3.3.1.1.3 attached to the appropriate pins. For test purposes, external electrical power sources shall supply 31.5 ± 0.5 VDC, -31.5 ± 0.5 VDC, +15.0 ± 0.1 VDC and -15.0 VDC, the -15.0 VDC supply to be the negative of +15.0 VDC supply within ± 0.04 VDC. CAUTION: Do not apply electrical power without applying helium to the control section per 3.3.1.1.4.

4.5.3.4 Second environmental sensors test. The SES, as installed on the control section, shall be subjected to changing magnetic fields as described in 3.3.1.4.1 and 3.3.1.4.2. Under these conditions, the SES Logic 1 and Logic 2 shall meet the output "go" requirements of 3.3.1.4.1 and the "no go" requirements of 3.3.1.4.2. Subsequent to "no go" testing, a waiting period of 30 minutes must be observed before further testing can begin. The changing magnetic field applied to each SES shall be supplied by a pulse generator which, when applied to the calibrating circuit of Figure 3, shall produce an output voltage rising to 16 ± 2 volts in 80 plus ten (10) minus zero (0) μs and decay to less than 10 volts in a total (rise and fall) pulse length of 95 ± 10 μs.

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4.5.3.5 Wing operation test.

4.5.3.5.1 Manual operation test. After installation of the wings within the control housing and prior to installation of the wing lock, the wing shall be adjusted to the specified sweep angle in accordance with drawing 9308000. The wings shall rotate freely in and out of the control housing.

4.5.3.5.2 Squib operation test. This test shall be conducted in conjunction with 4.5.3.6.5 through 4.5.3.6.7 as part of an AUR live sequence test. With the application of power to the wing unlock squib and the wing extend squib, the wings shall unlock and deploy, respectively. The position of each wing when extended and locked shall comply with the requirements of 3.3.1.2.2. The wings shall rotate freely out of the control housing.

4.5.3.5.3 Wing release test. This test shall be performed prior to the installation of the pressure cartridge and in accordance with drawing 9308100. A pressure of 500 psi maximum, using nitrogen, shall be applied to the pressure cartridge port. This pressure shall unlock the wings (i.e. the piston shall retract and the rotor shall rotate fully).

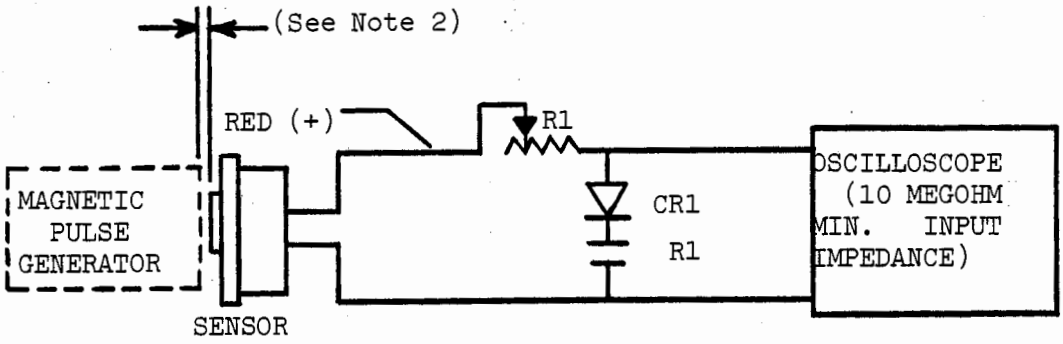
4.5.3.5.4 Wing extension test. This test shall be performed prior to the installation of the CAS and the wing release mechanism, and in accordance with drawing 9308000. The control section shall be in a vertical position and with the wings stowed and the wing tips pointing down. A pressure of 300 psi maximum, using nitrogen, shall be applied to plug port. The wings shall fully extend and locks in the extended position. The pressure shall be removed and the wings and the spider shall remain locked.

4.5.3.6 Fin operation test.

4.5.3.6.1 Fin lock test. With the fins installed on the control actuator subassembly and the subassembly installed in the control section housing, verify that the fin retention springs are engaged in the fin slots by applying a load of 1.5 $\pm$ 0.2 lbs in the fin notch outboard and normal to the fin rotational axis. The fins shall not deploy.

4.5.3.6.2 Fin extension test. With the control actuator subassembly installed in the control section housing and the fins bonded, release the tang of the fin retention spring from the fin notch. Care must be taken not to bend the springs or to slide the assembly aft. Place the control section in a vertical position and extend each of the fins by applying a load of less than or equal to 0.1 lb to each fin notch outboard and normal to the fin rotational axis. Each fin shall deploy. When fully extended the fin positions shall meet the deployed angle requirements of 3.3.1.3.1.

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SENSOR: SES per MIL-F-63230(AR)except that it shall be selected and maintained such that its Inductance shall be 1.53±0.05 millihenrys and its Resistance when added to R1 shall be 8.00±0.2 ohms.

R1: Calibrating Resistor (not to exceed 0.5 ohms).

CR1: JANIN645 Diode (or equivalent).

C1: Capacitor, 1.5± 10% ufd.

NOTE: 1) Discharge C1 prior to each test.
2) gap, if any, shall be the same for calibration as for test.

FIGURE 3 Calibration circuit for magnetic pulse generator

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4.5.3.6.3 Input signal/fin phasing test. Using external power and pneumatics (4.5.3.3), the control section fins shall rotate in response to input voltages to the pitch, roll, and yaw command channels as indicated by Table I. CAUTION: Do not conduct test without applying pneumatic power to the control section.

4.5.3.6.4 Closed loop position gain test. Using external power and pneumatics (4.5.3.3), command each actuator with a 0.02 ± 0.01 Hz sine wave of sufficient amplitude to move the actuators to generate feedback voltages in accordance with the rated values given in Figure 1 in pitch, yaw, and roll. Fin position feedback voltages relative to input command voltages shall be within the limits of Figure 1.

4.5.3.6.5 Shaft unlock/duration cycle test. This test shall be conducted with the control section electrically assembled into an AUR configuration (4.5.3). With each fin extended, locked and commanded in accordance with 3.3.1.3.2, apply control section battery power to activate the guidance section. The +30 VDC power shall be obtained from the control section battery. The +11 VDC power shall be supplied to the projectile from an external source using the special wiring provided by 4.5.1.6. Event sequencing shall be from normal guidance section operation (See MIL-P-63224). The elapsed time from application of the cutter initiation signal to achievement of output shaft command position, within the limits of Figure 1, shall be 200 ms or less. At 1 ± 0.5 second after application of cutter initiation signal, command each actuator with a 1 ± 0.1 Hz sine wave of sufficient amplitude to produce ±5 ±0.5 degrees deflection of the fins in pitch and yaw. (Deflection may be determined from feedback voltage values.) The time for gas container pressure to decrease to the downstream regulator pressure shall be as required by 3.3.1.3.5. Fin command signals shall be supplied to the control section from an external source. These tests shall be conducted in conjunction with the guidance section duty cycle test (MIL-P-63224).

4.5.3.6.6 Fin commands tests. This test shall be conducted with the control section electrically assembled into an AUR configuration. With each fin extended and locked, activate the guidance section by applying power from the control section battery. Event sequencing shall be from normal guidance section operation (see MIL-P-63224). The guidance section shall be used to generate pitch and yaw command signals. Fin position feedback voltages relative to input command voltages shall be within the limits of Figure 1. These tests shall be conducted in conjunction with guidance section fin command polarity test (MIL-P-63224).

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4.5.3.6.7 Functional test. This test shall be conducted with the control section electrically assembled into an AUR configuration. With each fin extended and locked, activate the guidance section by applying power from the control section battery. Event sequencing shall be from normal guidance section operation (see MIL-P-63224). The guidance section shall be used to generate pitch, yaw and roll commands. With the guidance section boresighted, roll aligned and in the "guidance enabled" mode, the roll command shall be -1.5 ± 0.4 volts. Position of fins 1 and 3 shall be as indicated by Table I. As the guidance section illumination is moved off boresight in pitch and yaw, the fin positions shall be as indicated by Table I. These tests shall be conducted in conjunction with the guidance section "Guidance Code setting 211 Test" and the "Guidance Code Setting 788 Test" (MIL-P-63224).

4.5.4 Environmental tests. The control section shall meet the performance requirements as specified in each of the following environmental methods as part of the test. During pre-environmental performance tests, extreme care shall be taken to ensure that the wing, battery, and gas squibs are not expended. The control sections subjected to environmental testing shall be configured as required by 3.3.2.

4.5.4.1 Temperature shock test. The unit under test (UUT) shall be subjected to the temperature shock tests of MIL-STD-810, Method 503.1, Procedure I with the exception that the low temperature of Step 2 shall be -50° ±2.5°F (-45.6 ±1.4°C), and step 9 shall be omitted.

4.5.4.2 Transportation vibration test. The UUT shall be subjected to the vibration tests of MIL-STD-810, Method 514.2, Procedure X using the criteria of Table 514.2 VII and Figure 514.2-7 curve AX except search and dwell tests shall be omitted. The UUT shall be securely fastened to the vibration table.

4.5.4.3 Fragility shock test. The UUT shall be subjected to the shock tests of MIL-STD-810, Method 516.2, Procedure I, using the shock pulse of Figure 516.2-2 with the exception that the peak acceleration value, 'A', shall be 200 g's and the nominal duration, 'D', shall be 10 ms. After exposure the UUT shall be disassembled and the control section shall be examined and show no evidence of physical damage as defined by 3.3.2 and shall meet the requirements of 4.5.3.1 through 4.5.3.4 and 4.5.3.6.1 through 4.5.3.6.4.

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4.5.4.4 Launch shock high temperature test. After exposure to the environments of 4.5.4.1 through 4.5.4.3 and after passing the tests of 4.5.3.1 through 4.5.3.4 and 4.5.3.6.1 through 4.5.3.6.4, the control section shall be installed in an appropriate canister and the canistered assembly exposed to an acceleration of 9000 ±300g, at 145 ±2.5°F (62.8 ± 1.4°C), in the forward direction by launching from an 8-inch gun, M115 or equivalent (GFE), within 5 minutes after removal from temperature conditioning. After exposure, the control section shall be examined and show no evidence of physical damage as defined by 3.3.2, and shall meet the performance requirements of 4.5.3.1 through 4.5.3.4 and 4.5.3.6.1.through 4.5.3.6.4.

4.5.4.5 Launch shock low temperature test. This test shall be identical to the test of 4.5.4.4 with the exceptions that the temperature shall be -25 ± 2.5°F (-31.7 ± 1.4°C) and the acceleration level shall be 8000 ± 400 g. After exposure, the control section shall be examined and shall show no evidence of physical damage as defined by 3.3.2 and shall meet the performance requirements of 4.5.3.1 through 4.5.3.4 and 4.5.3.6.1 through 4.5.3.6.4.

4.5.4.6 High and low temperature tests. The control section shall be electrically assembled into an AUR configuration (4.5.3) and shall meet the performance requirements of 4.5.3.6.5 through 4.5.3.6.7 while stabilized at temperatures of -25°F ± 2.5°F and +145°F ± 2.5°F.

4.5.4.7 Production vibration test. The control section shall be exposed to the random vibrations as required by 3.3.2.5. The fins shall be locked in the extended position. During exposure, without external fin loading, command each actuator with 0.02 ± 0.01 Hz sine wave of sufficient amplitude to move the actuators to ± 16.0 ± 1.0 degrees in pitch and ± 21 ± 1.0 degrees in yaw as determined from feedback voltages. The command voltages vs. feedback voltage shall be evaluated for one cycle of operation in each of the two planes. The feedback voltage relative to input command voltage shall fall within the limits of Figure 1. During the remainder of the vibration period the control actuator shall be exercised as follows with no recording required. Command the actuators to an offset of 15 ± 1 degrees in roll/yaw and 10 ± 1 degrees in pitch as determined from the feedback voltages. Supply a sine wave to each actuator over a frequency range of 0.7 to 20 Hz of sufficient amplitude as determined by feedback voltages, to produce ±2.5 ± 0.5 degrees of output motion about the offset position. The sweep time for the frequency excursion from 0.7 to 20 to 0.7 Hz shall be 30 ± 3.0 seconds. After exposure to these environments the control section shall meet the requirements of

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4.5.3.1, 4.5.3.2, 4.5.3.4, 4.5.3.6.2 through 4.5.3.6.4. At the conclusion of testing, the fins shall be manually rotated and locked into the control housing.

4.5.5 Workmanship inspection. The control section shall be examined to verify conformance with the workmanship requirements of 3.5.

4.5.6 Magnetic particle inspection. Magnetic particle shall be in accordance with MIL-I-6868 for the continuous wet, fluorescent method. MIL-I-6870 shall not apply. The entire wing release mount assembly shall be inspected twice, once after inducing a magnetic field while placed on a coil as shown in Figure 4 and once after inducing a magnetic field while placed in a coil as shown in Figure 5. Adequacy of the inspection process shall be checked as a minimum at the start and end of daily production using approved standards. During the check all defects shall be clearly defined. If the inspection process is determined to be inadequate, the condition(s) shall be corrected and all wing release mount assemblies inspected since the last check shall be reinspected. Parts examined by magnetic particle and found acceptable shall be identified with the letter "M" in such a manner as to have no deleterious effect on the part. The marking shall withstand normal handling.

4.5.6.1 Reduced inspection. If a random sample of 80 wing release mount assemblies are inspected per para. 4.5.6, and there are no defects found when inspected as shown in Figure 5 that were not evidenced when inspected as shown in Figure 4, then the inspection as shown in Figure 5 shall be required only at six month intervals (for the sample of 80) thereafter. However, if additional defects are evidenced using the orientation of Figure 5, then paragraph 4.5.6 shall apply until notification is given by the contracting officer that the reduced inspection may be reinstated.

4.5.7 Preparation for delivery inspection. The preparation for delivery of the item shall be inspected for compliance with the packing and packaging requirements of the procurement document (see 6.2(e)).

5. PACKAGING

5.1 General. Packaging and packing shall be level A, B or C, as specified (See 6.2(e)).

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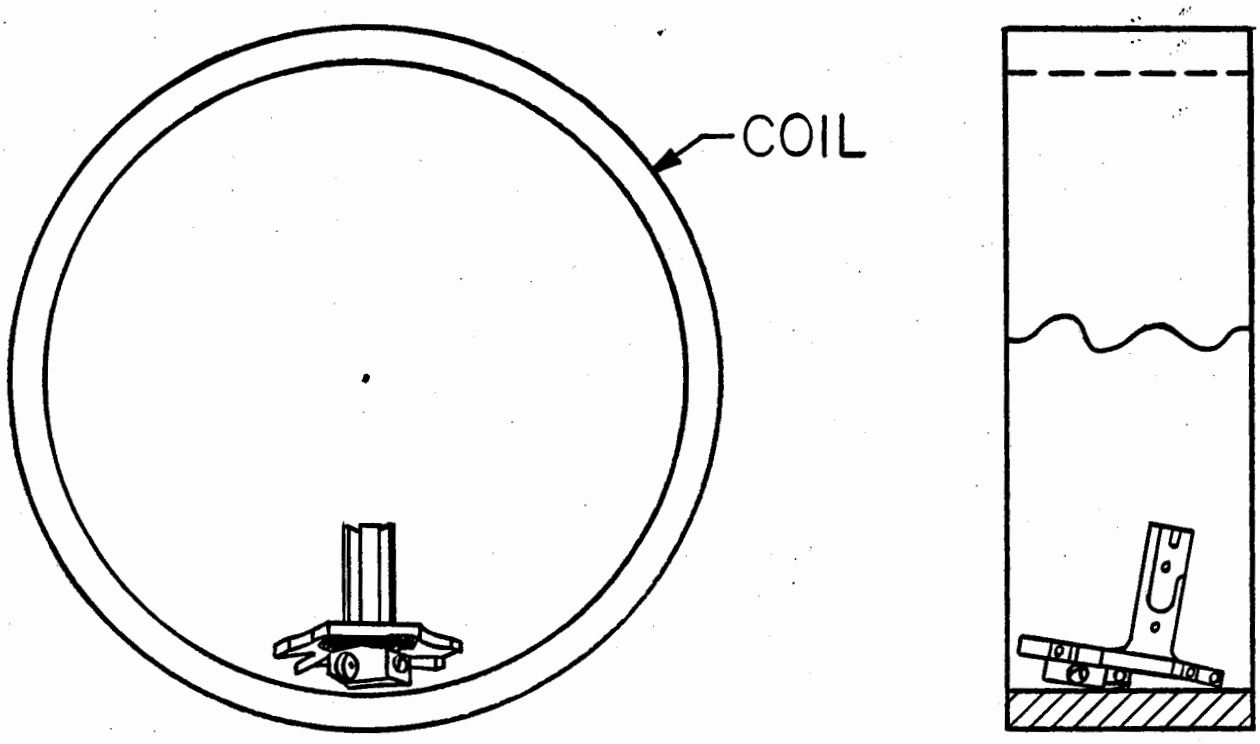


FIGURE 4 Magnetic particle inspection orientation

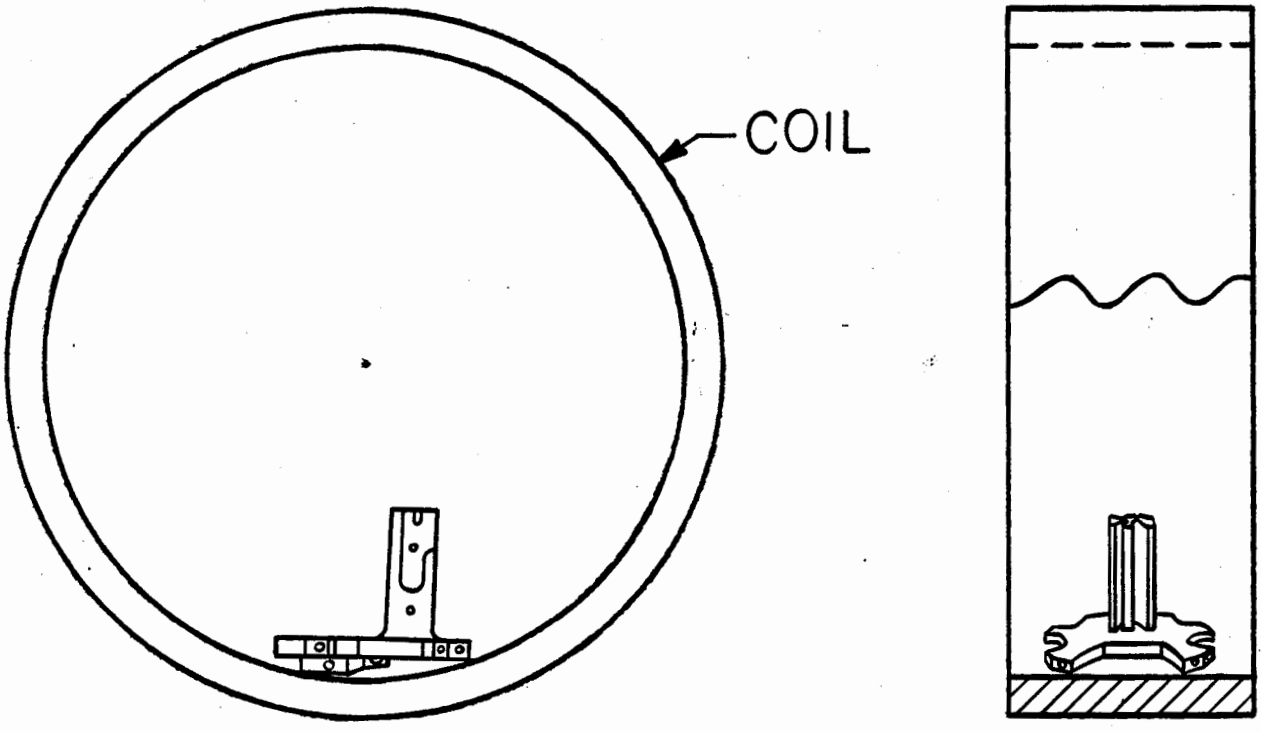


FIGURE 5 Magnetic particle inspection orientation

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5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be accomplished by wiping the exterior surfaces of the control section with a dry lint-free cloth. The internal portions of the assembly may be cleaned with prepared compressed air.

5.1.1.2 Packaging - Level A. Sharp edges and protrusions shall be covered with PPP-C-843 cushioning material. The assembly shall then be wrapped with Type II, MIL-P-17667, barrier material. Desiccant conforming to MIL-D-3464 shall be spaced uniformly throughout the assembly and secured with PPP-T-76 tape. The assembly shall then be heat-sealed with a Class F, MIL-B-117 bag.

5.1.1.3 Packing. The packaged control section shall be cushioned within the exterior container with polystyrene or polyethylene end pad and saddles. The exterior container shall conform to PPP-B-601 or PPP-B-621, overseas requirement.

5.1.2 Level B.

5.1.2.1 Cleaning. See 5.1.1.1

5.1.2.2 Packaging - Level B. Sharp edges and protrusions shall be cushioned with PPP-C-843. The assembly shall then be heat sealed within a Class F, MIL-B-117 bag.

5.1.2.3 Packing. Packing shall be accomplished as defined in 5.1.1.3 with the exception that the exterior container shall be a domestic type.

5.1.3 Level C.

5.1.3.1 Cleaning. See 5.1.1.1

5.1.3.2 Packaging. The control section shall be packaged in a manner that will prevent contamination of the item from the supply source to the first receiving activity for immediate use.

5.1.3.3 Packing. The control section, either single unit or multiples thereof, shall be properly secured in a strong wooden or metal box. The gross weight of a box containing one or more assemblies shall not exceed 250 pounds.

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5.2 Marking. In addition to the standard unit and exterior markings required in MIL-STD-129, each exterior container shall be marked "ELECTRIC SQUIB" and have affixed thereto a class C explosive label, and a nonflammable gas label.

5.3 Transportation. Electric squibs are acceptable for commercial air shipments, and military air shipments.

6. NOTES

6.1 Intended use. The control section is intended for use as a section of the M712 Projectile.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Quantity required and delivery schedules.

(c) Serialization requirements, if applicable.

(d) Quality Conformance Inspection, if other than specified in Section 4 of this specification.

(e) First Article Sample requirements, if other than specified in Section 4 of this specification.

(f) Packaging requirements, if other than specified in Section 5 of this specification.

(g) Certificates of conformance for each lot or shipment of product.

(h) Requirements for lot control.

6.2.2 Contract data requirements.

(a) Contract data requirements for inspection equipment designs (conforming to data Item description DI-R-1714, Tailored.)

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(b) Data Cards. When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item description DI-L-1410, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Definitions.

6.3.1 Contractor. Unless otherwise specified, the word contractor used throughout this specification shall mean any supplier or producer of items or material to the Government Owned, Contractor Operated (GOCO) plants.

6.3.2 Technical data package (TDP). A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists of specifications, standards, models, performance requirements, quality assurance provisions, and packaging data. Clarification of Quality Assurance terms and definitions are available in MIL-STD-109.

6.4 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in section 4, or on the equipment lists referenced on the equipment tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE designs. SIE designs may consist of any of the following:

(a) Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

(b) A source control drawing or a specification control drawing as defined in MIL-STD-100.

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(c) An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When the envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered.

6.4.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. All submissions shall be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAR-I(D), Dover, NJ 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities in now under the cognizance of ARDC.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Preparing Activity:
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(Project 1320-A796)

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